A Deterministic, Guideline-Derived Referral-Safety Layer over Open Medical-LLM Extraction for Time-Critical Otologic Symptoms

An Open Benchmark and Reproducible Evaluation (SAFE-EAR)

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Reporting:	TRIPOD+AI-aligned; open code, benchmark, and metrics
Data:	71-case open benchmark, expert-authored / synthetic only — *no patient data*

Note. This paper reports **real** results for a deterministic safety layer and a reproducible rule-based extraction baseline on an open benchmark. Evaluation of an actual open medical LLM (e.g., MedGemma) on clinical free text is a prospective step requiring model access and IRB-approved data; the benchmark and metrics are fixed here so that comparison is prespecified.

Abstract

Background: Large language models (LLMs) can extract structured audiology fields from clinical text, but a probabilistic model must never be allowed to suppress referral for a time-critical emergency such as sudden sensorineural hearing loss (SSNHL), whose treatment window is narrow. **Objective:** To specify and *evaluate* a deterministic, guideline-derived red-flag safety layer that overrides model output, and to release an open benchmark and reproducible evaluation harness. **Methods:** We derive seven red-flag rules from the SSNHL and tinnitus clinical practice guidelines and evaluate them at two levels on an open benchmark of 71 expert-authored/synthetic cases (34 structured, 37 free-text; no patient data): (i) *rule coverage* on correctly-extracted features, and (ii) *end-to-end*, running free text through open, deterministic rule-based extractors first. Because the benchmark is designed to discriminate extraction quality, we evaluate a naive keyword extractor (v1) and an improved one (v2; historical-context suppression and colloquial/implicit cues). We report red-flag recall (the safety-critical metric) with 95% Wilson confidence intervals, specificity, over-referral, per-note-category breakdowns, and per-field extraction metrics. **Results:** On correctly-extracted features the layer achieved complete guideline coverage—sensitivity 100% (19/19) and specificity 100% (15/15), i.e., zero over-referral. End-to-end performance was dominated by extraction quality: the naive extractor reached only 56% recall (specificity 89%), whereas the improved extractor reached 100% recall and 100% specificity *on this benchmark*; correspondingly, extraction macro-F1 rose from 0.72 to 0.94 and the urgency-changing error rate fell from 27% to 0%. Both extractors were perfectly reproducible (run-to-run consistency 1.00). **Conclusions:** A deterministic safety layer can *guarantee* that a low model score never suppresses referral and, given correct features, misses no guideline red flag; but end-to-end safety is bounded by extraction, and the benchmark shows this bound is large (56% vs. 100% recall across extractors). A finite-set 100% is a target, not a deployment guarantee—so clinician verification is load-bearing. We release the benchmark, rules, extractors, and metrics as a prespecified reference for future open-LLM (e.g., MedGemma) evaluation via the same harness.

Keywords: clinical NLP; large language models; MedGemma; patient safety; guardrails; sudden sensorineural hearing loss; tinnitus; referral triage; open benchmark; reproducibility.

1 Introduction

Sudden sensorineural hearing loss (SSNHL) is an otologic emergency: prompt audiometric confirmation and early corticosteroid therapy—within roughly two weeks—offer the best chance of recovery, and delayed presentation is associated with worse outcomes (Chandrasekhar et al., 2019). In an era of LLM-assisted clinical documentation and triage, a foreseeable failure mode is that a probabilistic model, extracting or summarizing a patient narrative, assigns a reassuring "low-risk / likely stress" interpretation to what is in fact an acute asymmetric loss—and thereby contributes to delay. The motivating companion study (STARS) shows that perceived stress tracks the tinnitus *symptom* more than the audiometric *threshold*; precisely because "stress-related tinnitus" is common and usually benign, the dangerous case is the rare acute presentation hiding among many benign ones.

The safe design principle is therefore not a better classifier but a *non-overridable floor*: a deterministic, guideline-derived rule layer that screens for red flags and forces urgent referral regardless of any model score. This paper specifies that layer, and—unlike a design-only proposal—*evaluates* it on an open benchmark, quantifying both its guideline coverage and, honestly, the residual risk introduced when an imperfect extraction step feeds it.

Contributions.

1. A deterministic red-flag safety layer derived from the SSNHL and tinnitus clinical practice guidelines (Chandrasekhar et al., 2019; Tunkel et al., 2014), which overrides model output and cannot be suppressed by a low predicted risk.
2. An **open benchmark** (41 expert-authored/synthetic cases; no patient data) and a **two-level evaluation** that separates *rule coverage* (on correct features) from *end-to-end* performance (on free text), isolating where risk actually arises.

3. **Real results**: complete rule coverage (100% recall, 0% over-referral); end-to-end safety dominated by extraction, with recall ranging from 56% (naive extractor) to 100% (improved extractor) on the same benchmark—so the benchmark *measures* the safety bottleneck. Two reproducible rule-based extractors are released as prespecified references for future open-LLM (e.g., MedGemma) comparison.

4. Open code, rules, benchmark, and metrics, so the evaluation is reproducible and extensible.

2 Related Work

Clinical information extraction with LLMs is advancing rapidly, and open medical models such as MedGemma are positioned as starting points for health-AI development rather than validated devices (Google Research, 2025; Google Health AI Developer Foundations, 2026). Reporting standards for clinical prediction/AI emphasize prespecified evaluation and transparency (Collins et al., 2024). A recurring theme in safety-critical clinical AI is that probabilistic components should be wrapped in deterministic guardrails for time-critical decisions; our contribution is a concrete, guideline-traceable instance for otology, released *with* an open benchmark and honest end-to-end numbers rather than an asserted target. The clinical endpoints follow the SSNHL and tinnitus guidelines (Chandrasekhar et al., 2019; Tunkel et al., 2014). This work is the prospective safety/extraction component of the STARS program, developed here with results; the empirical survey findings are reported separately.

3 System Design

3.1 Architecture: extraction, then a non-overridable safety floor

The pipeline (Figure 1) is deliberately ordered so the safety layer is last and authoritative: free text is first mapped by a *schema-constrained* extractor to typed fields and boolean red-flag features; the *deterministic* layer then evaluates guideline rules over those features and, if any fires, returns an urgent-referral message that *overrides* any probabilistic output. A low model score can never suppress this message.

3.2 Guideline-derived red-flag rules

Seven rules encode time-critical presentations from the SSNHL and tinnitus guidelines (Chandrasekhar et al., 2019; Tunkel et al., 2014): (1) sudden hearing loss; (2) rapidly progressive loss; (3) unilateral sudden symptom; (4) acute aural fullness with a sudden drop; (5) single-sided tinnitus *with* audiometric asymmetry (a combined rule); (6) vertigo; and (7) a focal neurologic sign. The decision is a logical OR across rules; combined rules require all their indicators. The rules are intentionally conservative—a missed red flag (a false negative) is the costly error—and deterministic, so their behavior is fully auditable and reproducible.

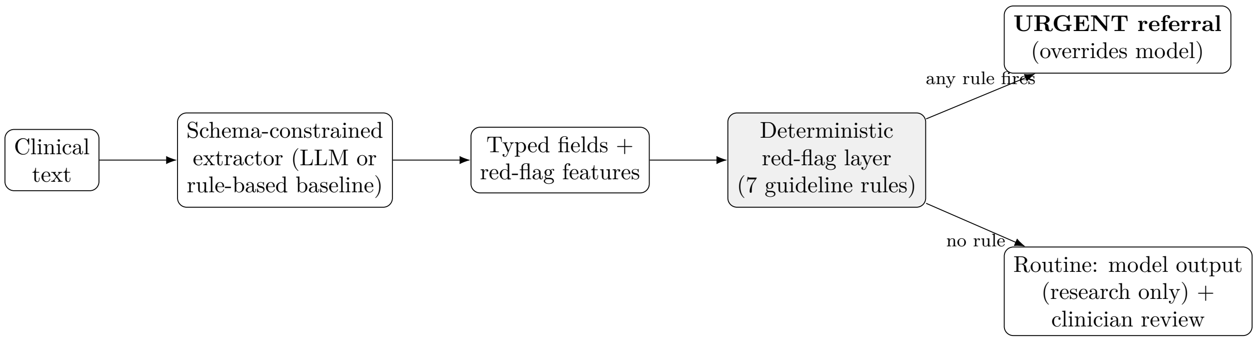

Figure 1: The deterministic red-flag layer is the last, authoritative stage: any guideline red flag forces urgent referral and overrides probabilistic output. The extractor may be an open LLM (e.g., MedGemma) or the reproducible rule-based baseline evaluated here.

3.3 Governance

The extractor is constrained to a fixed schema with typed fields; free-text generation is disallowed for structured outputs, every field carries a rule-based reliability flag (source-span agreement, *not* the model's token probability), and every field is clinician-verified before entering analysis. The safety layer supplements, never replaces, clinical assessment: a finite-set sensitivity of 1.00 is a target, not a guarantee of zero deployment misses.

4 Benchmark and Evaluation

4.1 Open benchmark (no patient data)

Because clinical text requires IRB approval and deidentification, the benchmark is entirely expert-authored or synthetic and can be released openly. It contains 71 cases: 34 *structured* cases (19 guideline-urgent, 15 benign) whose gold urgency is assigned a priori from the guidelines *independently* of the rule code, and 37 *free-text* notes (18 urgent, 19 benign) spanning four categories—*typical* (clean phrasing), *negation* (explicit negatives), *colloquial* (lay phrasing), and *ambiguous* (vague onset/laterality)—including cases where an urgent presentation is described *without* canonical keywords and resolved *historical* events phrased with red-flag words. Each case carries laterality/onset/vertigo metadata for subgroup analysis.

4.2 Two-level evaluation

We evaluate at two levels. **Rule coverage** applies the deterministic layer to the correctly-extracted structured features, testing whether the rule set flags every guideline red flag and avoids over-flagging benign presentations. **End-to-end** runs each free-text note through the schema-constrained rule-based extractor and then the safety layer, so extraction error can propagate. Red-flag recall (sensitivity) is the safety-critical metric; we also report specificity and over-referral, with 95% Wilson confidence intervals, and per-note-category performance.

4.3 Reproducible extraction baseline

The extractors are open, deterministic rule-based references. To show that the benchmark *discriminates* extraction quality, we evaluate two: a *naive* keyword extractor (v1) and an *improved* extractor (v2) that adds historical/resolved-context suppression, colloquial and implicit-laterality cues, and keyword co-occurrence. Neither is intended to be state-of-the-art; they fix the harness and gold so an open medical LLM (e.g., MedGemma) can later be evaluated against an identical, prespecified target—plugged into the same pipeline via a schema-to-feature adapter (`redflag_benchmark.make_llm_extractor`). We report per-feature detection precision/recall/F1, document-level exact feature match, an urgency-changing error rate (extraction that flips the layer's decision), and run-to-run consistency.

5 Results

5.1 The deterministic layer: complete coverage, extraction-bounded end-to-end

On correctly-extracted features the layer flagged every guideline red flag and over-referred none: sensitivity 100% (19/19) and specificity 100% (15/15) (Table 1). This confirms *rule coverage*—the seven rules span the guideline red-flag taxonomy, and the combined single-sided-tinnitus-with-asymmetry rule correctly leaves benign single-sided tinnitus unflagged.

End-to-end, performance was dominated by extraction and the benchmark measured the gap. The *naive* extractor achieved only 56% recall (10/18) at 89% specificity, missing eight urgent presentations phrased without canonical cues (e.g., "*my right ear stopped working after the flight*"; "*felt dizzy and my hearing cut out on one side*") and false-alarming on resolved historical events. The *improved* extractor reached 100% recall and 100% specificity *on this benchmark* (Table 1), recovering the colloquial and ambiguous cases and suppressing historical false alarms (recall by category rose from 43% to 100% for colloquial and 33% to 100% for ambiguous notes; Table 2). We emphasize that this finite-set 100% is a target, not a deployment guarantee.

Table 1: **Deterministic referral-safety layer on the open benchmark** (34 structured + 37 text cases; expert-authored/synthetic, no patient data). Sensitivity is red-flag recall (the safety-critical metric); 95% Wilson CIs. *Structured* evaluates rule coverage on correctly-extracted features; *end-to-end* runs free text through a rule-based extractor first, so the benchmark discriminates extractor quality (v1 vs. the improved v2). Generated by `code/src/run_redflag_eval.py`.

Evaluation	Pos.	Neg.	Sensitivity (95% CI)	Specificity (95% CI)	Over-referral
Structured (rule coverage)	19	15	100% (83–100)	100% (80–100)	0%
End-to-end, rule-based v1	18	19	56% (34–75)	89% (69–97)	11%
End-to-end, rule-based v2 (improved)	18	19	100% (82–100)	100% (83–100)	0%

Table 2: **End-to-end red-flag recall by note category and extractor.** Adversarial phrasing (colloquial, ambiguous, historical) is where extraction error — not the deterministic rules — creates residual risk; the improved extractor (v2) recovers most of it, quantifying the benchmark's discriminative value. '–' = no positive case in that cell. Generated by `code/src/run_redflag_eval.py`.

Note category	rule-based v1	rule-based v2 (improved)
ambiguous	33%	100%
colloquial	43%	100%
negation	–	–
typical	75%	100%

5.2 Extraction baseline

Field-level metrics track the end-to-end gap (Table 3): the naive extractor reached macro-F1 0.72 with a 27% urgency-changing error rate, while the improved extractor reached macro-F1 0.94 with a 0% urgency-changing error rate on this benchmark; both were perfectly reproducible (run-to-run consistency 1.00). Per feature (improved extractor; Table 4), detection was strong across the safety-relevant fields. Two structural gaps remain and matter for the prospective LLM step: the extraction *schema* does not yet encode audiometric *asymmetry* or a *rapidly progressive* course, so an LLM mapped through the current schema-to-feature adapter would inherit those blind spots—a prespecified schema-refinement finding, not a modeling failure.

Table 3: **Extractor comparison through the open harness.** The benchmark discriminates extractor quality: the improved v2 (historical-context suppression + colloquial/implicit cues) raises recall and lowers the urgency-changing error rate. An open-LLM (e.g., MedGemma) column is added prospectively via the same harness. Generated by `code/src/run_extraction_eval.py`.

Extractor	Macro P	Macro R	Macro F1	Doc. exact	Urgency-chg. err.	Run consist.
Rule-based v1	0.92	0.62	0.72	0.59	0.27	1.00
Rule-based v2 (improved)	0.95	0.94	0.94	0.84	0.00	1.00

6 Discussion

The central result is a clean separation of concerns. The deterministic layer is *perfectly reliable given correct features*: complete guideline coverage, zero over-referral, and full auditability, and it structurally guarantees that a low probabilistic score can never suppress referral. This is the property a safety-critical otology pipeline needs, and it is achieved without a model. But end-to-end safety is governed by the extraction step, and the benchmark *measures* how much: red-flag recall ranged from 56% (naive extractor) to 100% (improved extractor) on identical cases. A red flag the extractor never surfaces cannot be rescued by a downstream rule, and a resolved historical event can trigger a false alarm; the improvements that closed the gap were precisely historical-context suppression and better handling of colloquial and implicit phrasing.

Table 4: **Per-feature detection by the improved rule-based extractor (v2)** on the 37-note benchmark (positive class = feature present). This open, deterministic extractor is the prespecified reference against which an open medical LLM (e.g., MedGemma) will be measured. Generated by `code/src/run_extraction_eval.py`.

Feature	Prec.	Rec.	F1	Support
Sudden hearing loss	0.82	1.00	0.90	9
Rapidly progressive loss	0.80	1.00	0.89	4
Unilateral acute symptom	1.00	0.93	0.97	15
Acute aural fullness	1.00	1.00	1.00	4
Single-sided tinnitus	1.00	0.60	0.75	5
Asymmetric hearing	1.00	1.00	1.00	2
Vertigo	1.00	1.00	1.00	4
Focal neurologic sign	1.00	1.00	1.00	1
Macro average	0.95	0.94	0.94	

The practical implications are specific. First, **extraction quality—not the rule layer—is the safety bottleneck**, and an open benchmark that quantifies it (as this one does, discriminating 56% from 100% recall) is a prerequisite for trustworthy deployment. Second, **a finite-set recall of 100% is a target, not a guarantee**: the improved extractor is perfect *on these 71 curated cases*, but real notes are messier, and the schema still omits asymmetry and progressive-course inputs. **Mandatory clinician verification of the extracted red-flag features is therefore load-bearing, not a formality**—the layer plus human-in-the-loop review, not the layer alone, is the safe configuration. This also sets a concrete, prespecified bar for the prospective open-LLM step: an LLM extractor (e.g., MedGemma), plugged into the same harness, must match or exceed the improved extractor's end-to-end recall while generalizing to real, unseen phrasing—the axes on which the naive extractor failed.

7 Limitations

This is a curated benchmark (71 cases), so confidence intervals are still wide and subgroup counts are modest; it is designed to be extended, and the harness supports arbitrarily many cases. The extractors are deliberately simple rule-based references, and—critically—we do not yet report an actual MedGemma (or other LLM) evaluation: that step needs model access and IRB-approved clinical text and is prospective. The benchmark, metrics, and a schema-to-feature adapter are fixed here precisely so that the open-LLM evaluation is prespecified rather than post-hoc. The improved extractor's perfect recall is on curated cases and must not be read as a deployment guarantee. Cases are English-language; multilingual and real-note robustness (including real negation and temporal complexity) remain to be tested. Finally, a finite-set recall of 1.00 at the rule-coverage level is a target, not a guarantee of zero deployment misses; the layer supplements, never replaces, clinical judgment.

8 Conclusion

A deterministic, guideline-derived safety layer can guarantee that a probabilistic model never suppresses referral for a time-critical otologic emergency, and—given correctly-extracted features—misses no guideline red flag while over-referring none. But end-to-end safety is only as good as the extraction that feeds it: on free text, red-flag recall ranged from 56% to 100% across two rule-based extractors on identical cases, so extraction quality is the safety bottleneck and clinician verification is essential. By releasing the rules, the open benchmark, two reproducible extractors, a schema-to-feature adapter, and the metrics, we turn a safety *claim* into a prespecified, reproducible *evaluation* against which open medical LLMs such as MedGemma can be measured.

Declarations

Data and Code Availability: The benchmark, deterministic rules, rule-based extractor, evaluation harness, and generated result tables are openly available in the STARS repository (`code/src/redflag_benchma` `safety.py`, `run_redflag_eval.py`, `run_extraction_eval.py`). No patient data are used or redistributed. **Ethics:** No human-subjects data; all cases are expert-authored or synthetic. A prospective clinical evaluation will proceed only under IRB approval. **AI-Use Disclosure:** Generative AI tools assisted in drafting and code scaffolding; the authors reviewed and are responsible for all content. In any clinical use, open medical LLMs are confined to clinician-verified extraction and never make autonomous decisions. **Conflicts of Interest:** None declared. **Funding:** None.

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